UCS3461 Foundations of Artificial Intelligence

NATURE OF ENVIRONMENTS

Semester 4, B.E. Computer Science and Engineering
15 December 2025

Sakaya Milton R
SSN College of Engineering

What is AI?

- Artificial intelligence is the analysis and synthesis of computational agents that act intelligently.
- An agent is something that acts in an environment.
- A computational agent perceives, reasons, and acts by computing.

Rational Agents

- Science of making machines that

	Human performance	Ideal performance
Thought process	Think like human	Think rationally
Behaviour	Act like human	Act rationally

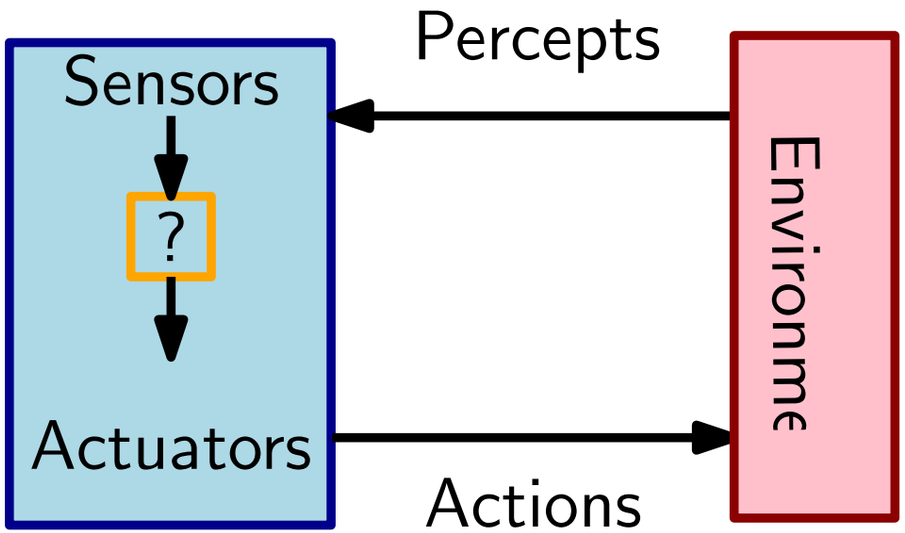
- Intelligent agents $\equiv$ Rational agents

Computational Rationality

- Act rationally
- Goals expressed as the utility of outcomes
- Rational: maximally achieve the goals
- Rational $\equiv$ Maximize Your Expected Utility
- Rationality only concerns
  - what decisions are made
  - not the thought process behind them
- General (mathematical and engineering) principles of rational agents and on components for constructing them.

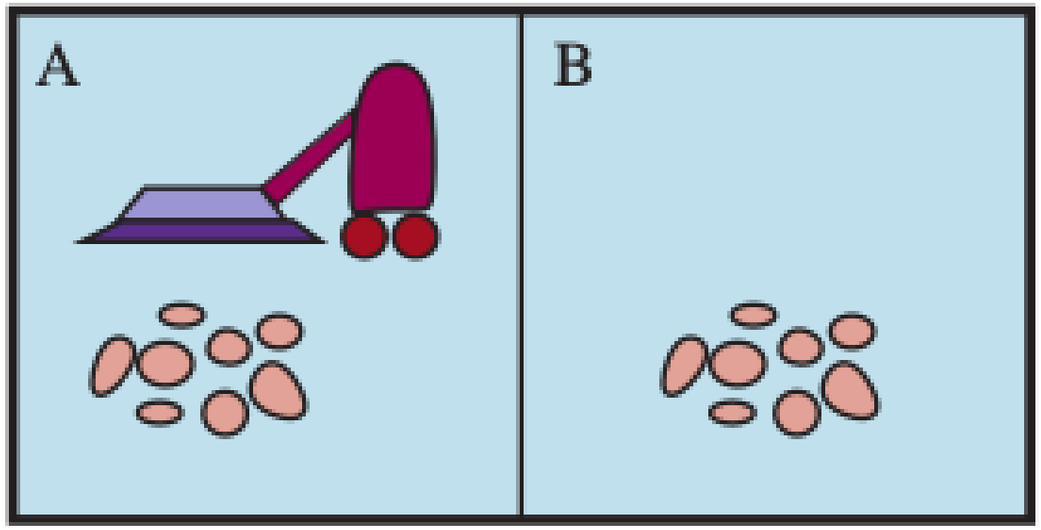
Agents and Environments

- An agent perceives its environment through sensors and acts upon it through actuators
- A rational agent chooses actions to maximize the expected utility
- Percept: agent's perceptual inputs at any given instant.
- Agent function describes an agent's behavior – maps every possible percept sequence to an action
- Agent program implements agent function.



Example: Vacuum Cleaner Agent

- Percepts
  - Location: left, right
  - Cleanness: dirty, clean
- Action
  - Suck
  - Move left
  - Move right
- Agent function

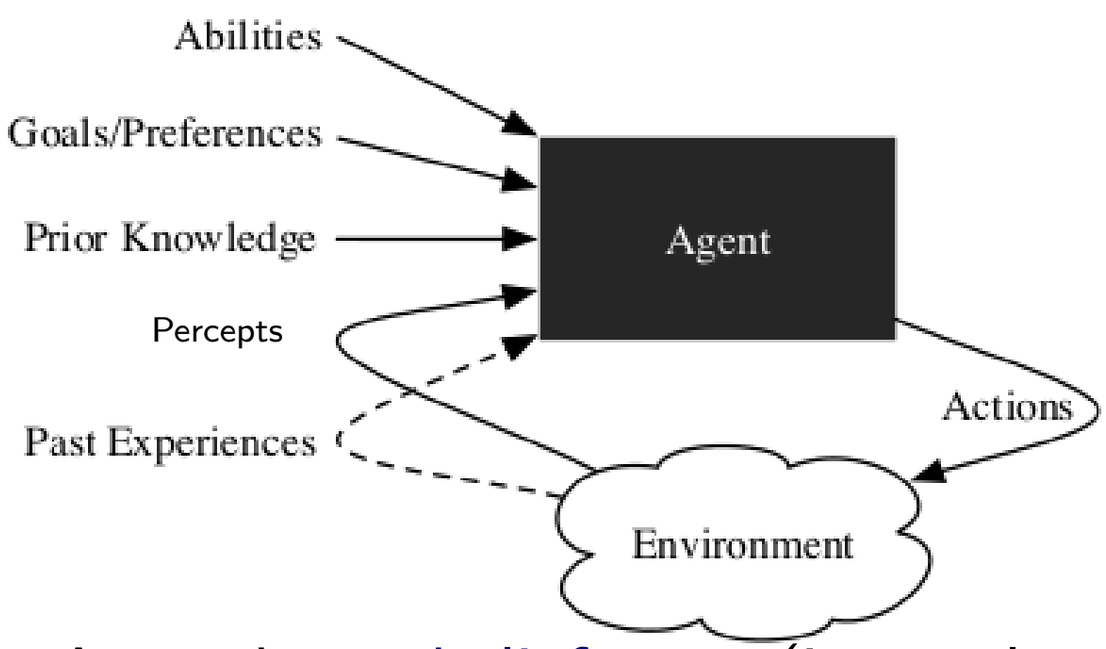


Percept sequence	Action
(A, clean)	right
(A, dirty)	suck
(B, clean)	left
(B, dirty)	suck
(A, clean), (A, clean)	right
(A, clean), (A, clean), (A, dirty)	suck

Agents Situated in Enviroment

- Agent: a coupling of perception, reasoning, and acting.
- Agent acts in an environment which may include other agents.
- Agent + environment = world

Agents Acting in an Environment: Inputs and Outputs



- Agent has a belief state (internal state) – encodes agent's belief about its environment.
- Updates the belief state based on percepts.
- Uses belief state and percept to decide its actions.

- Goals to achieve or preferences over states. . . .
- Prior Knowledge the world – it doesn't get it from experience, it knows it when it comes into being.
- History of percepts (percept sequence): percept/observation/stimuli it receives from environment now.
- Past experiences/data received in the past.
- Abilities: the set of possible actions it can perform

Rationality

1. Agent's prior knowledge of the environment.
2. Agent's percept sequence to date.
3. Actions that the agent perform.
4. Performance measure defines the criterion of success.

For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Example Agent: Thermostat for Heater

- goals/preferences: conformable temperature, save fuel, save money
- prior knowledge: 24 hour cycle, weekends
- percepts: temperature, set temperature, who is home, outside temperature
- past experiences: when people come and go, who likes what temperature
- abilities: turn heater on or off

Example Agent: Autonomous Car

- goals/preferences: safety, get to destination, timeliness
- prior knowledge: street maps, what signs mean, what to stop for
- percepts: vision, laser, GPS, voice commands . . .
- past experiences: how braking and steering affects direction and speed
- abilities: steer, accelerate, brake

Example Agent: Robot

- goals/preferences: deliver food, rescue people, score goals, explore.
- prior knowledge: what is important feature, categories of objects, what a sensor tell us.
- percepts: vision, sonar, sound, speech recognition, gesture recognition.
- past experiences: effect of steering, slipperiness, how people move.
- abilities: movement, grippers, speech, facial expressions.

Nature of Environments

- Task environments – “problems”
- Rational agents – “solutions”

Specify the Task Environment

- PEAS
  - Performance
  - Environment
  - Actutators
  - Sensors
- Artificial environments defined by keyboard input and character output on a screen.
- Software agents (softbots)
- Internet as an environment
  - Complexity rivals that of the physical world
  - Inhabitants include many artificial and human agents

Examples

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers

Examples

Agent Type	Performance Measure	Environment	Actuators	Sensors
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts, bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard

Properties of Task Environment

Dimensions of complexity

- Modularity: flat, modular, hierarchical
- Time axis: discrete, continuous
- Planning horizon: episodic, sequential
- Representation: states, features, relations
- Computational limits: perfect rationality, bounded rationality
- Learning: knowledge is given, knowledge is learned
- Sensing uncertainty: fully observable, partially observable
- Effect uncertainty: deterministic, stochastic
- Preference: goals, complex preferences
- Number of agents: single agent, adversary, multiple agents
- Interaction: offline, online

Modularity

- Model at one level of abstraction: flat
- Model with interacting modules that can be understood separately: modular
- Model with modules that are (recursively) decomposed into modules: hierarchical
- Flat representations are adequate for simple systems.
- Complex biological systems, computer systems, organizations are all hierarchical
- A flat description is either continuous or discrete. Hierarchical reasoning is often a hybrid of continuous and discrete.

Time axis

When the state of the environment changes with time.

- Discrete: Percepts and actions of the agent at discrete time steps
- Continuous

Dynamism

- Dynamic: if the environment can change while an agent is deliberating – otherwise static.
- Semidynamic: if the environment itself does not change with time but the agent's performance score does.

Planning Horizon

How far the agent looks into the future when deciding what to do.

- Episodic: next episode does not depend on the actions taken in previous episodes. Agent need not think ahead.
- Sequential: current decision could affect all future decisions.
  - Finite stage: agent reasons about a fixed finite number of time steps
  - Indefinite stage: agent reasons about a finite, but not predetermined, number of time steps
  - Infinite stage: the agent plans for going on forever (process oriented)

Representation

Find compact representations of states and exploit the compactness for computational gains.

A state is one way the world could be. Agent reasons in terms of

- Explicit states: enumerate the states.
- Features or propositions.
  - States can be described using features.
  - 30 binary features can represent $2^{30} = 1,073,741,824$ states.
- Individuals and relations
  - There is a feature for each relationship on each tuple of individuals.
  - Often an agent can reason without knowing the individuals or when there are infinitely many individuals.

Computational Limits

- Perfect rationality: the agent can determine the best course of action, without taking into account its limited computational resources.
- Bounded rationality: the agent must make good decisions based on its perceptual, computational and memory limitations.

Learning from Experience/Data

- Refers to the agent's (or designer's) state of knowledge about the "laws of physics" of the environment.
- Whether the model is fully specified a priori:
  - Knowledge is given.
  - Knowledge is learned from data or past experience.
- Always some mix of prior (innate, programmed) knowledge and learning – nature vs nurture?
- Learning is impossible without prior knowledge (bias).

Prior Knowledge vs Learning from Experience

- Some initial knowledge as well as an ability to learn.
- After sufficient experience of its environment, the behavior of a rational agent can become effectively independent of its prior knowledge.
- Incorporation of learning allows one to design a single rational agent that will succeed in a vast variety of environments.

Uncertainty

There are two dimensions for uncertainty – sensing uncertainty and effect uncertainty. In each dimension an agent can have

- No uncertainty: the agent knows what is true
- Disjunctive uncertainty: there is a set of states that are possible
- Probabilistic uncertainty: a probability distribution over the worlds.

Sensing Uncertainty

Whether an agent can determine the state from its percepts:

- Fully observable: sensors detect complete state of the environment (relevant for action) at each time point.
- Partially observable: sensors miss parts of the state. There can be a number states that are possible given the agent's percepts.
- Unobservable: no sensors at all.

Effect Uncertainty

If an agent knew the current state and its action, could it predict the next state?

- Deterministic: the current state and the action taken by the agent completely determines the next state of the environment.
- Non-deterministic: different possible outcomes, but no probabilities
- Stochastic: for the same state-action, different next states possible – uncertainty about outcomes is quantified as probabilities.
- Uncertain: if not fully observable or stochastic

Preference

What does the agent try to achieve?

- Achievement goal is a goal to achieve. This can be a complex logical formula.
- Complex preferences may involve tradeoffs between various desiderata, perhaps at different times.
  - ordinal: only the order matters
  - cardinal: absolute values also matter

Examples: coffee delivery robot, medical doctor

Number of Agents

Single agent vs multiagent? Are there multiple reasoning agents that need to be taken into account?

- How to separate an agent and an environment?
- For an agent A, is entity B an agent or a part of the environment?
- B is an agent iff B tries to maximize a performance measure whose value depends on agent A's behavior.
- Single agent reasoning: any other agent is a part of the environment.
- Adversarial reasoning considers another agent, where when one agent wins, the other loses – two-player zero-sum game
- Multiple agent reasoning: an agent reasons strategically about the reasoning of other agents, perhaps needing to coordinate or cooperate.

Agents can have their own goals: cooperative, competitive, or goals can be independent of each other

Interaction

When does the agent reason to determine what to do?

- Reason offline: before acting
- Reason online: while interacting with environment

Two Extreme Cases

			Easiest			
Single	Fully	Deterministic	Episodic	Static	Discrete	Known
			Hardest			
Multi	Partial	Stochastic	Sequential	Dynamic	Continuous	Unknown

State-space Search

- Modularity: flat, modular, hierarchical
- Planning horizon: finite stage, indefinite stage, infinite stage
- Representation: states, features, relations
- Computational limits: perfect rationality, bounded rationality
- Learning: knowledge is given, knowledge is learned
- Sensing uncertainty: fully observable, partially observable
- Effect uncertainty: deterministic, stochastic
- Preference: goals, complex preferences
- Number of agents: single agent, adversary, multiple agents
- Interaction: offline, online

Deterministic Planning

- Modularity: flat, modular, hierarchical
- Planning horizon: finite stage, indefinite stage, infinite stage
- Representation: states, features, relations
- Computational limits: perfect rationality, bounded rationality
- Learning: knowledge is given, knowledge is learned
- Sensing uncertainty: fully observable, partially observable
- Effect uncertainty: deterministic, stochastic
- Preference: goals, complex preferences
- Number of agents: single agent, adversary, multiple agents
- Interaction: offline, online

Decision Networks

- Modularity: flat, modular, hierarchical
- Planning horizon: finite stage, indefinite stage, infinite stage
- Representation: states, features, relations
- Computational limits: perfect rationality, bounded rationality
- Learning: knowledge is given, knowledge is learned
- Sensing uncertainty: fully observable, partially observable
- Effect uncertainty: deterministic, stochastic
- Preference: goals, complex preferences
- Number of agents: single agent, adversary, multiple agents
- Interaction: offline, online

Markov Decision Processes (MDPs)

- Modularity: flat, modular, hierarchical
- Planning horizon: finite stage, indefinite stage, infinite stage
- Representation: states, features, relations
- Computational limits: perfect rationality, bounded rationality
- Learning: knowledge is given, knowledge is learned
- Sensing uncertainty: fully observable, partially observable
- Effect uncertainty: deterministic, stochastic
- Preference: goals, complex preferences
- Number of agents: single agent, adversary, multiple agents
- Interaction: offline, online

Reinforcement Learning

- Modularity: flat, modular, hierarchical
- Planning horizon: finite stage, indefinite stage, infinite stage
- Representation: states, features, relations
- Computational limits: perfect rationality, bounded rationality
- Learning: knowledge is given, knowledge is learned
- Sensing uncertainty: fully observable, partially observable
- Effect uncertainty: deterministic, stochastic
- Preference: goals, complex preferences
- Number of agents: single agent, adversary, multiple agents
- Interaction: offline, online

Classical Game Theory

- Modularity: flat, modular, hierarchical
- Planning horizon: non-planning, finite stage, indefinite stage, infinite stage
- Representation: states, features, relations
- Computational limits: perfect rationality, bounded rationality
- Learning: knowledge is given, knowledge is learned
- Sensing uncertainty: fully observable, partially observable
- Effect uncertainty: deterministic, stochastic
- Preference: goals, complex preferences
- Number of agents: single agent, adversary, multiple agents
- Interaction: offline, online

Humans

- Modularity: flat, modular, hierarchical
- Planning horizon: finite stage, indefinite stage, infinite stage
- Representation: states, features, relations
- Computational limits: perfect rationality, bounded rationality
- Learning: knowledge is given, knowledge is learned
- Sensing uncertainty: fully observable, partially observable
- Effect uncertainty: deterministic, stochastic
- Preference: goals, complex preferences
- Number of agents: single agent, adversary, multiple agents
- Interaction: offline, online

Examples

Task Environment					
Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle					
Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock					
Fully	Multi	Deterministic	Sequential	Semi Discrete	Discrete
Poker					
Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon					
Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving					
Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis					
Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis					
Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot					
Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller					
Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Interactive English tutor					
Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Summary

- Rational agents work in a task environment.
- Rational agents perceive the state of the environment, and takes an action through actuators, aiming to maximize the expected utility.
- Design (behaviour and structure) of rational agents depend on the nature their environment.
- What are the possible structures of agents?

End